Terraform-01 Task Documentation

Objective

The objective of this task is to install Terraform, understand core Terraform commands, execute sample Terraform templates shown in the training video, and integrate a sample Terraform project with Jenkins.

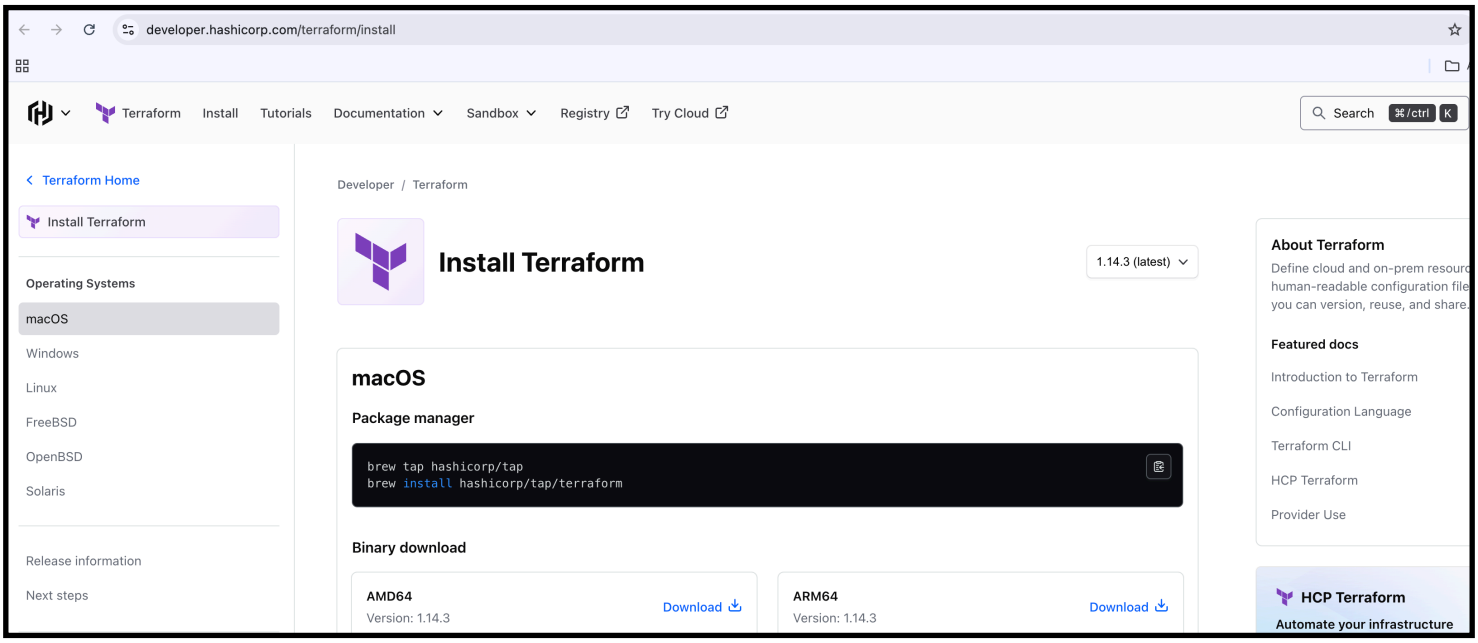
Task :1

1) Install Terraform on your PC

Step 1: Download

https://developer.hashicorp.com/terraform/downloads

Choose **macOS**



Now go to your local terminal add below commands

Step 2: Tap HashiCorp again (fresh metadata)

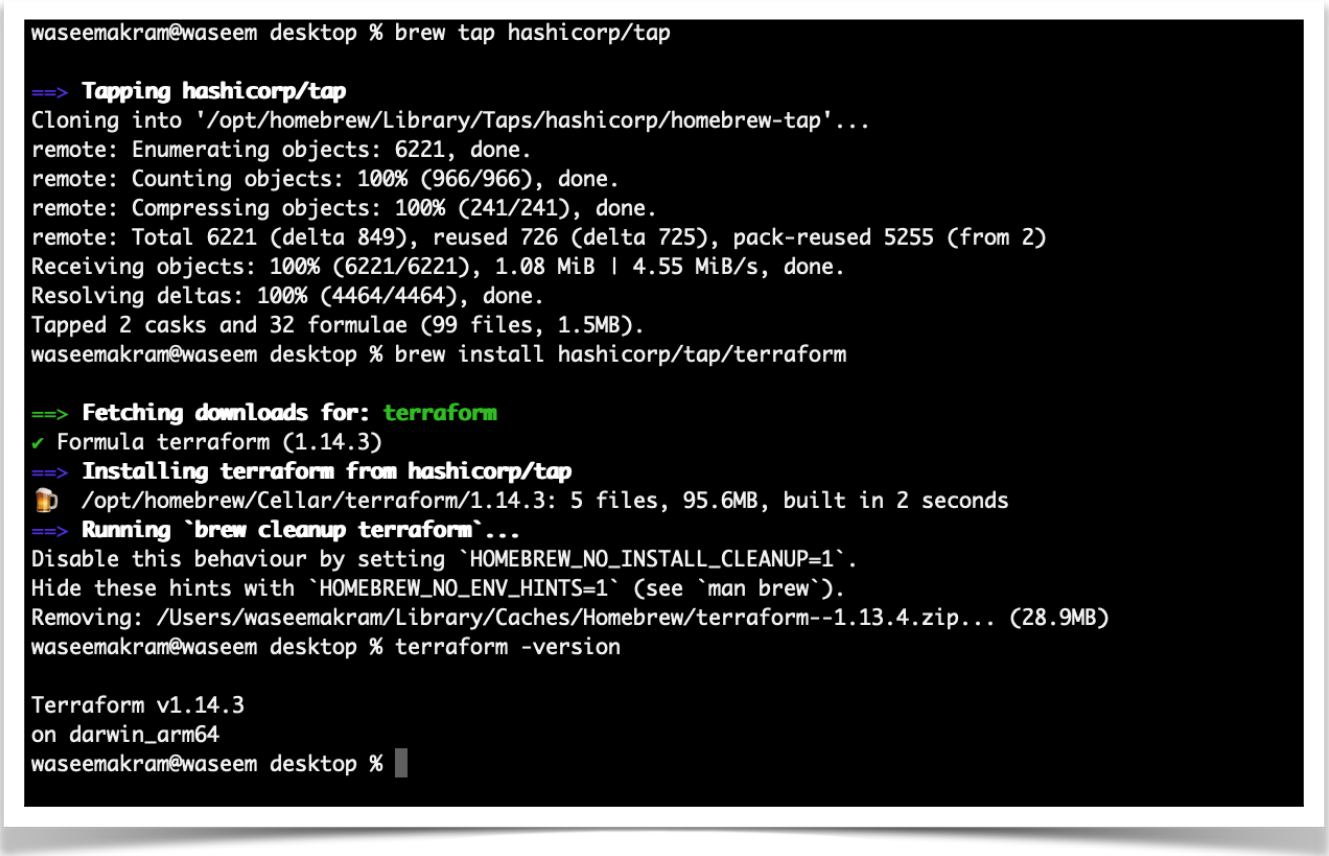
```
brew tap hashicorp/tap
```

🔷 **Step 3: Install latest Terraform**

```
brew install hashicorp/tap/terraform
```

🔷 **Step 4: Final Verification**

```
terraform -version
```

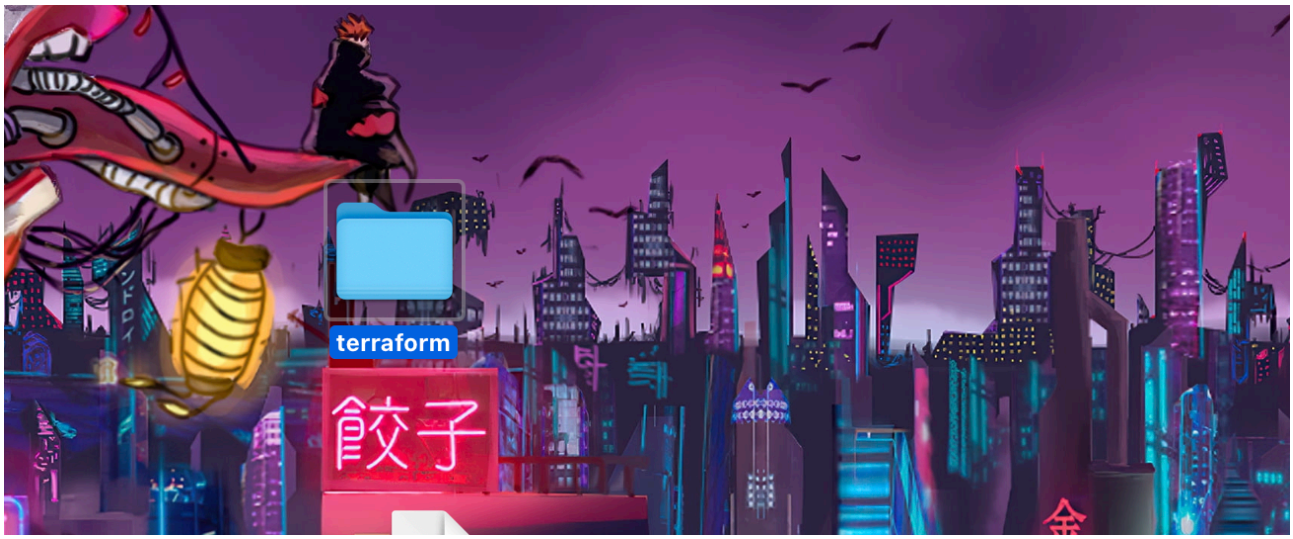


Task 2:

2). Execute all the templates shown in video.

Step 1: Create Your First Terraform Project

—>Make a new folder for your project (example: terraform).

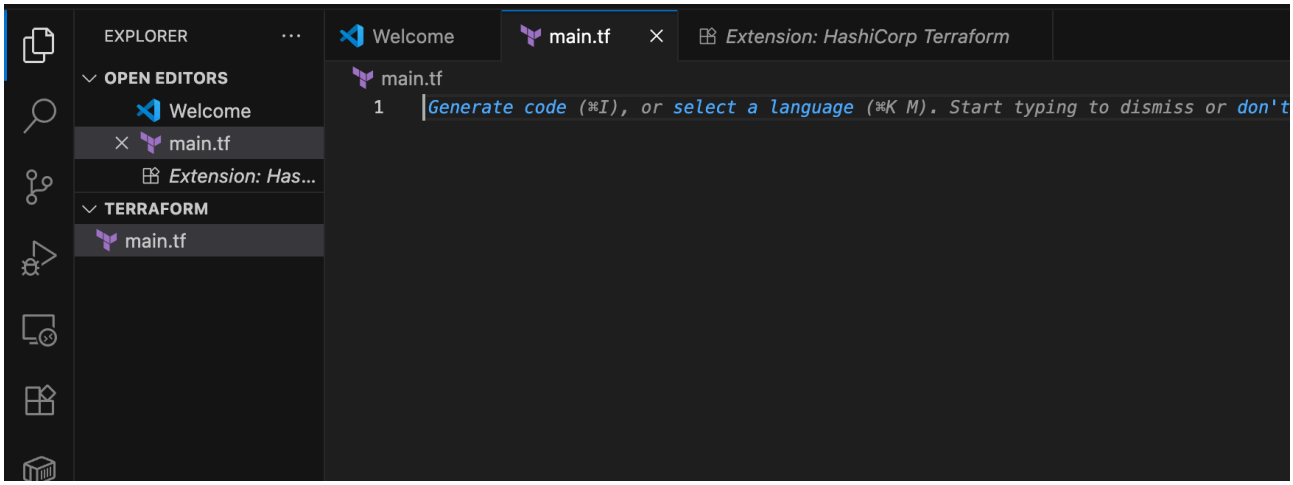


—->And open the folder in the vs code

—>Inside the folder , create your first Terraform file:

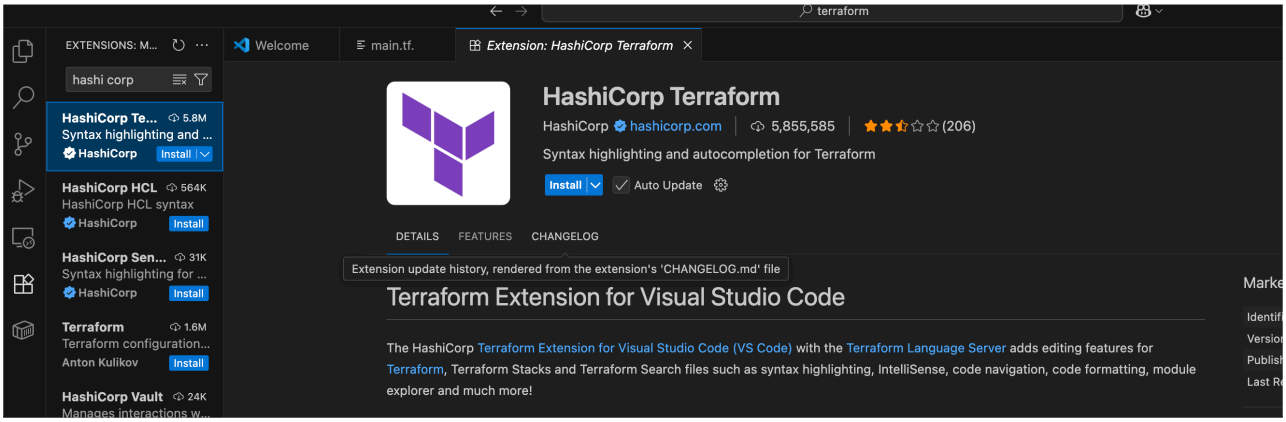
In VS Code → **File → New File → save as main.tf**.

Write a smaple content in it



—> .tf —>is the extension for the terraform resource files

—> Make sure you have hashi corp tool installed

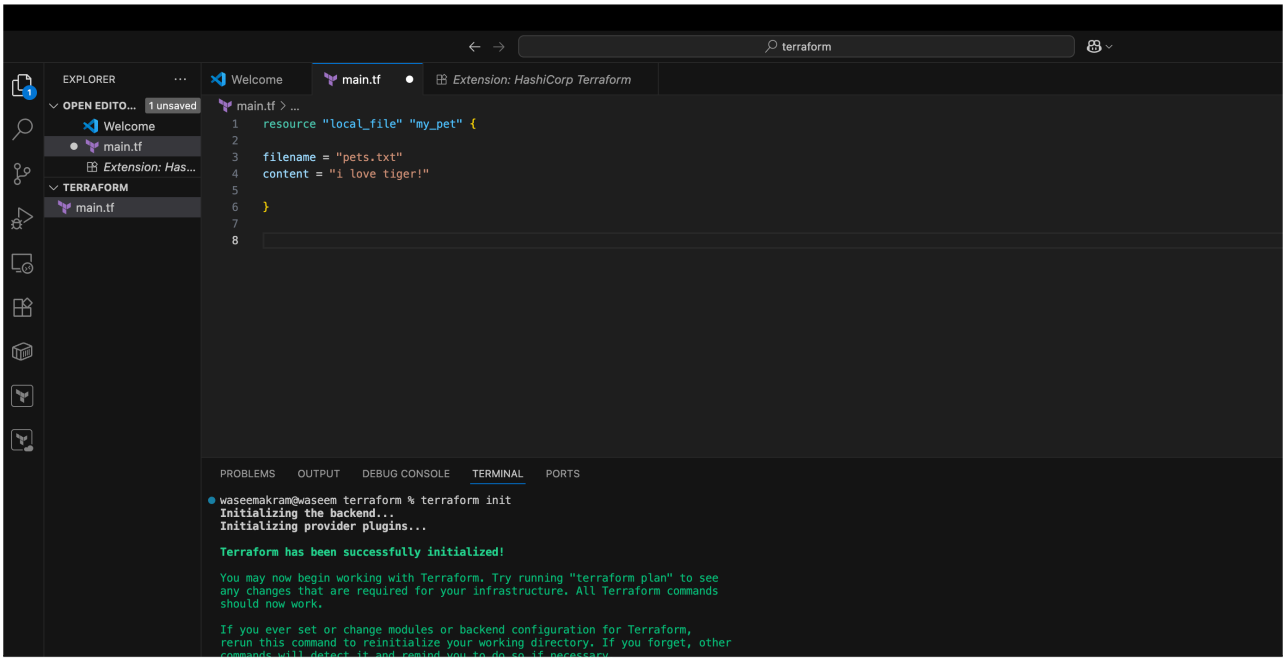


Install ✔

—> Write a smaple content in it

```
resource "local_file" "my_pet" {

filename = "pets.txt"
content = "i love tiger!"

}
```

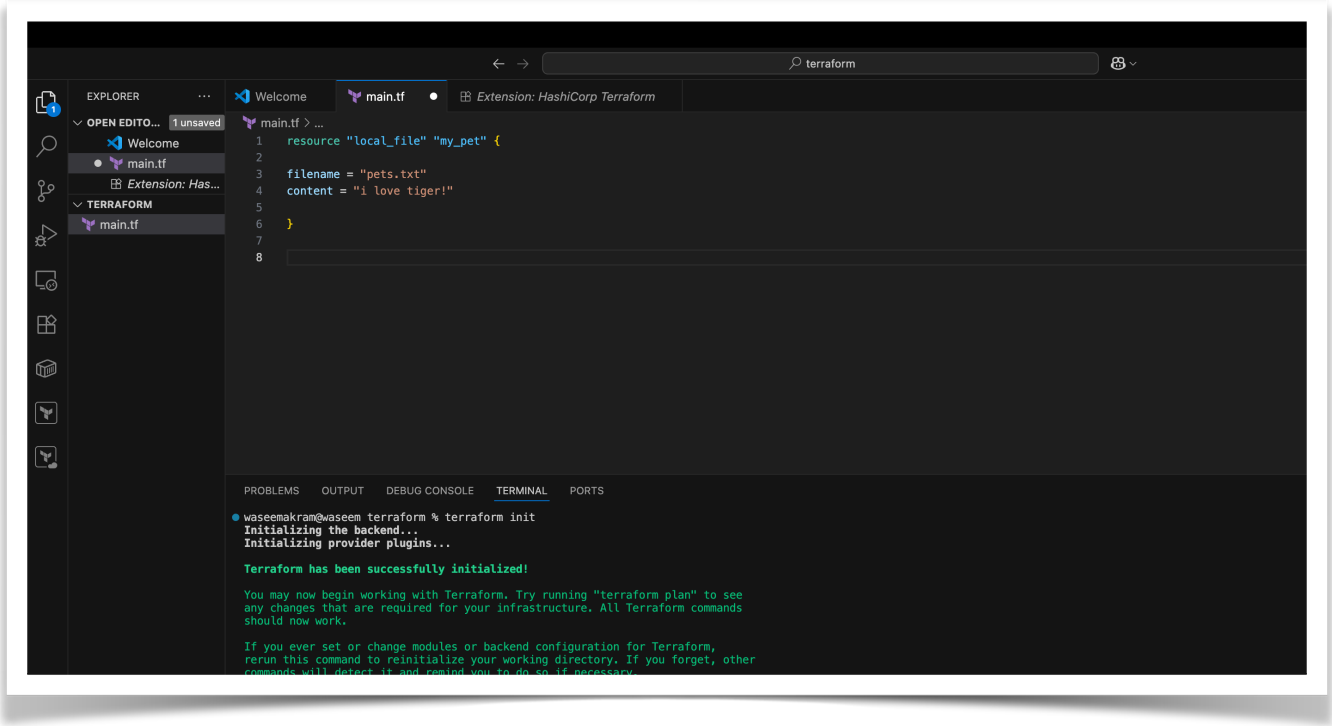
—>save



Step 2: Run Terraform Commands

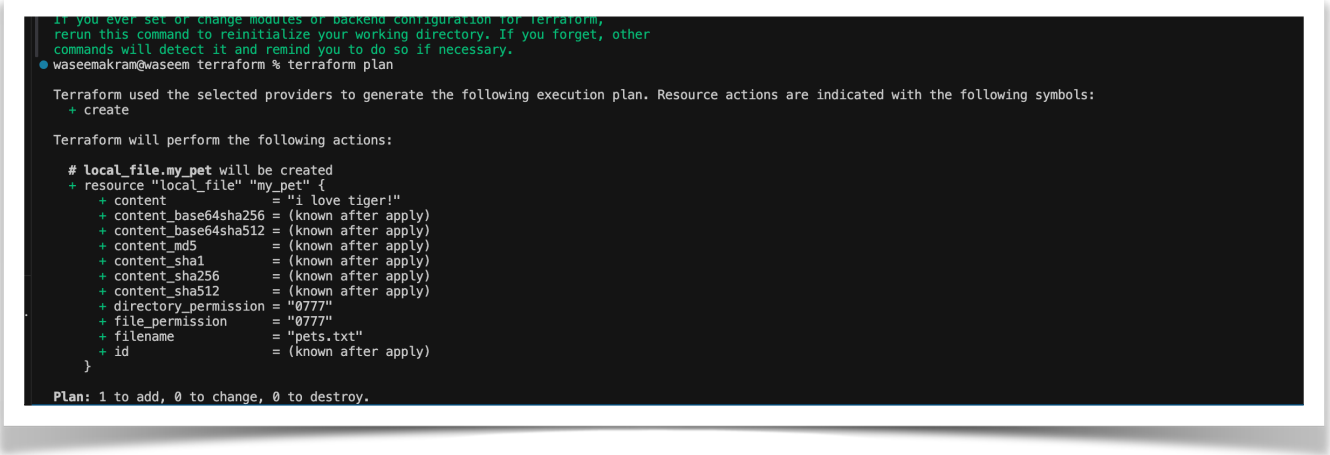
Initialize Terraform

terraform init



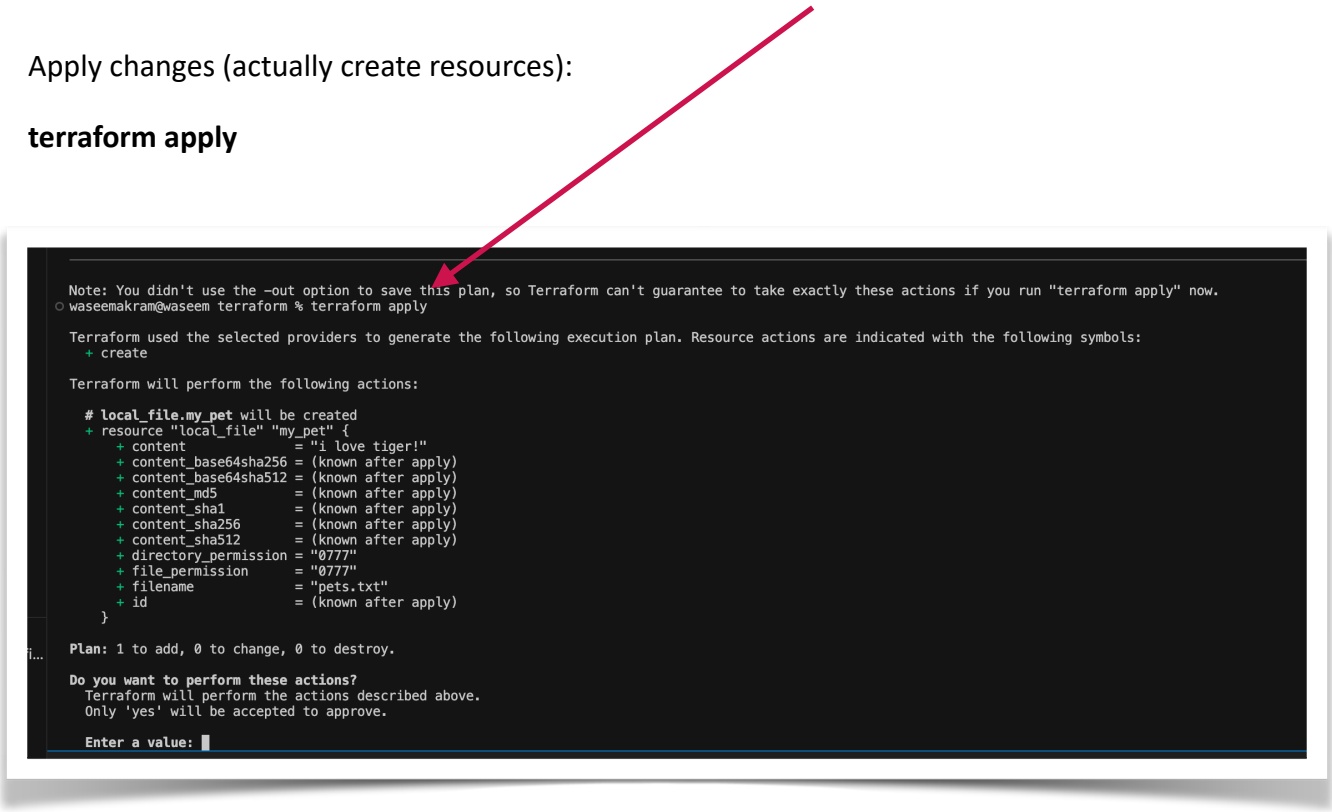
Now See what will be created:

terraform plan :

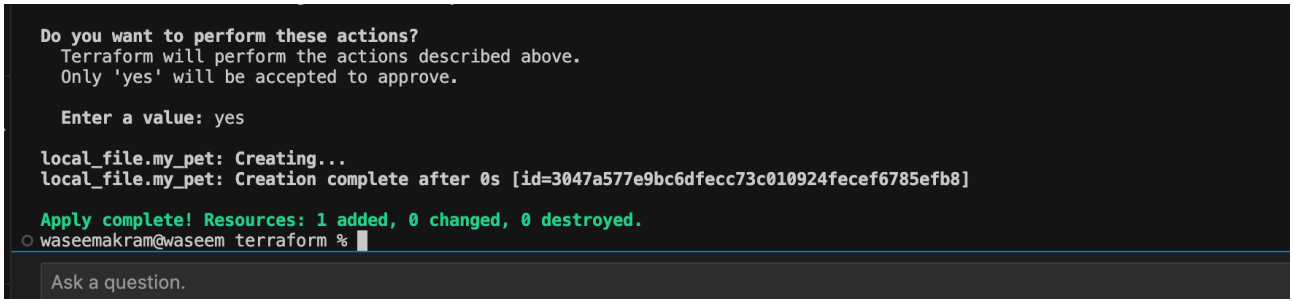


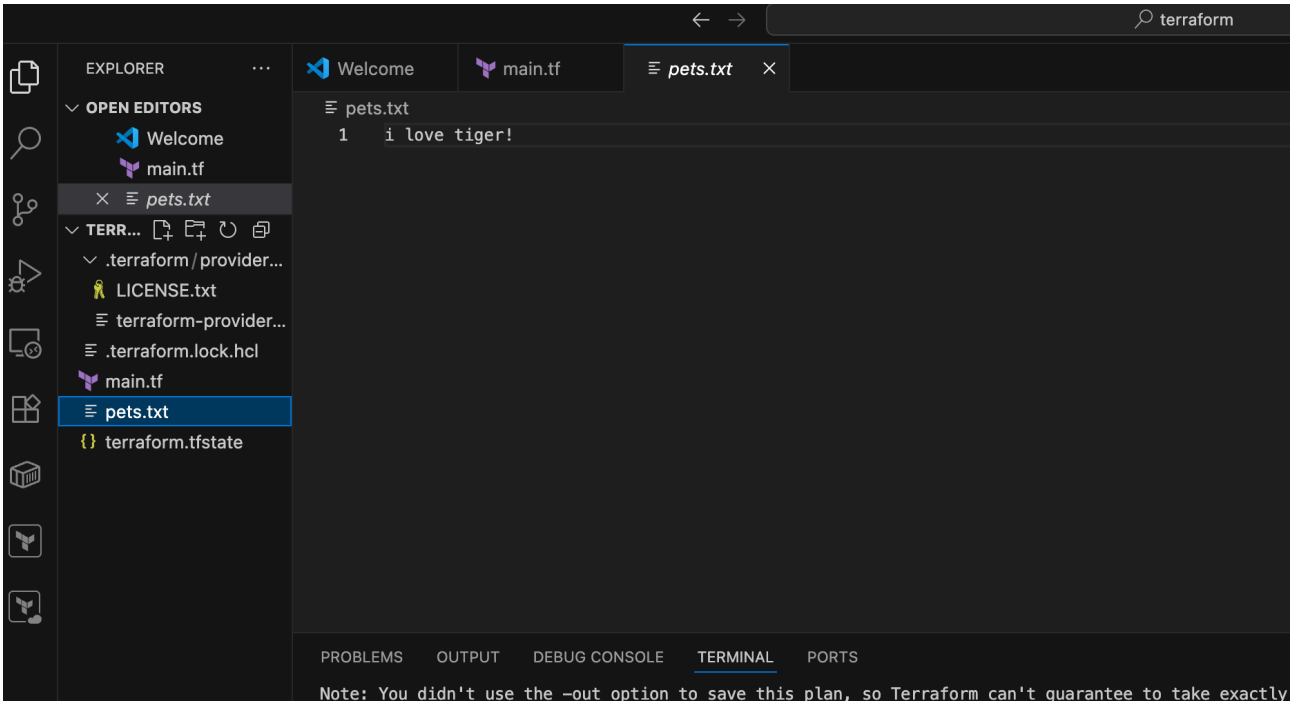
Apply changes (actually create resources):

terraform apply



Type `yes` when asked.





Destroy resources when finished:

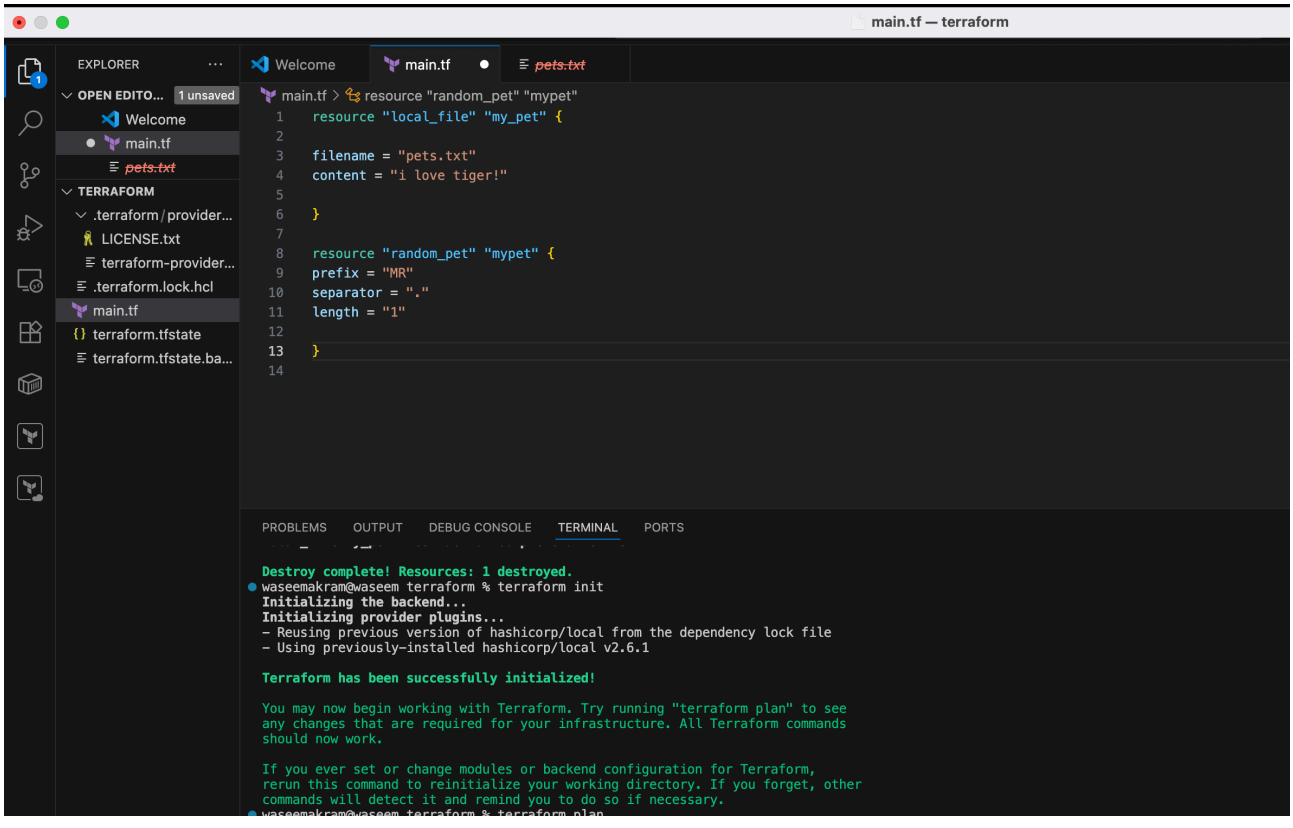
terraform destroy

```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
waseemakram@waseem terraform % terraform destroy
local_file.my_pet: Refreshing state... [id=3047a577e9bc6dfecc73c010924fecef6785efb8]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # local_file.my_pet will be destroyed
  - resource "local_file" "my_pet" {
      - content              = "i love tiger!" -> null
      - content_base64sha256 = "64NMjKd9RhYYRtpEW80nR0iV8XmTn/EQxjCsV2iQ8Rw=" -> null
      - content_base64sha512 = "04GiSpYFTyjzhvLQtI8OIDD92fJqNyRlNAHxZGbki7jOO49r+aOeEeUn3mgwkU80foQ6vyTMgXdajcJZ0YuWnw==" -> null
      - content_md5          = "1be4fcbf5b338961ca604c149be3dc22" -> null
      - content_sha1         = "3047a577e9bc6dfecc73c010924fecef6785efb8" -> null
      - content_sha256       = "eb834c8ca77d46161846da445bcd27474895f179939ff110c630ac576890f11c" -> null
      - content_sha512       = "d381a24a96054f28f386f2d0b48f0e2030fdd9f26a3724653401f16466e48bb8ce3b8f6bf9a39e11e527de6830914f347e843abf24cc81775a8
      - directory_permission = "0777" -> null
      - file_permission      = "0777" -> null
      - filename             = "pets.txt" -> null
      - id                   = "3047a577e9bc6dfecc73c010924fecef6785efb8" -> null
    }

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value:
```

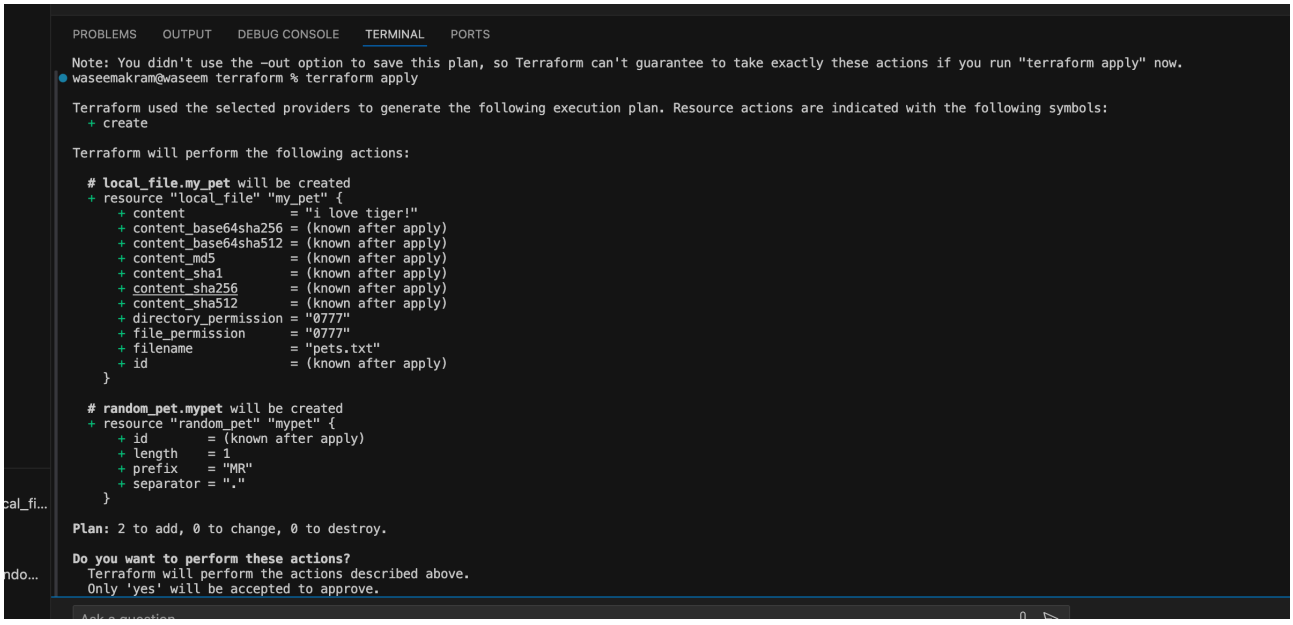
Now
—> created one more resource

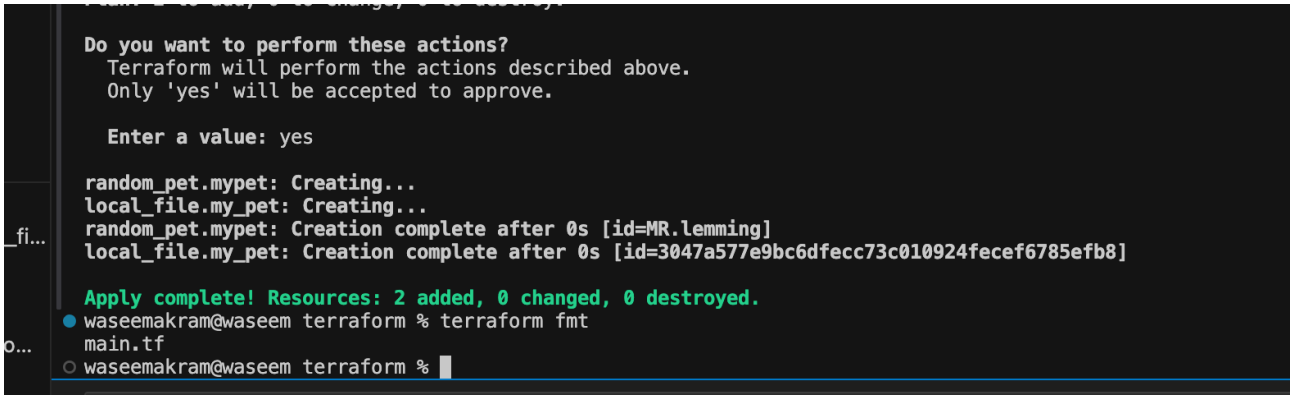
Terraform init

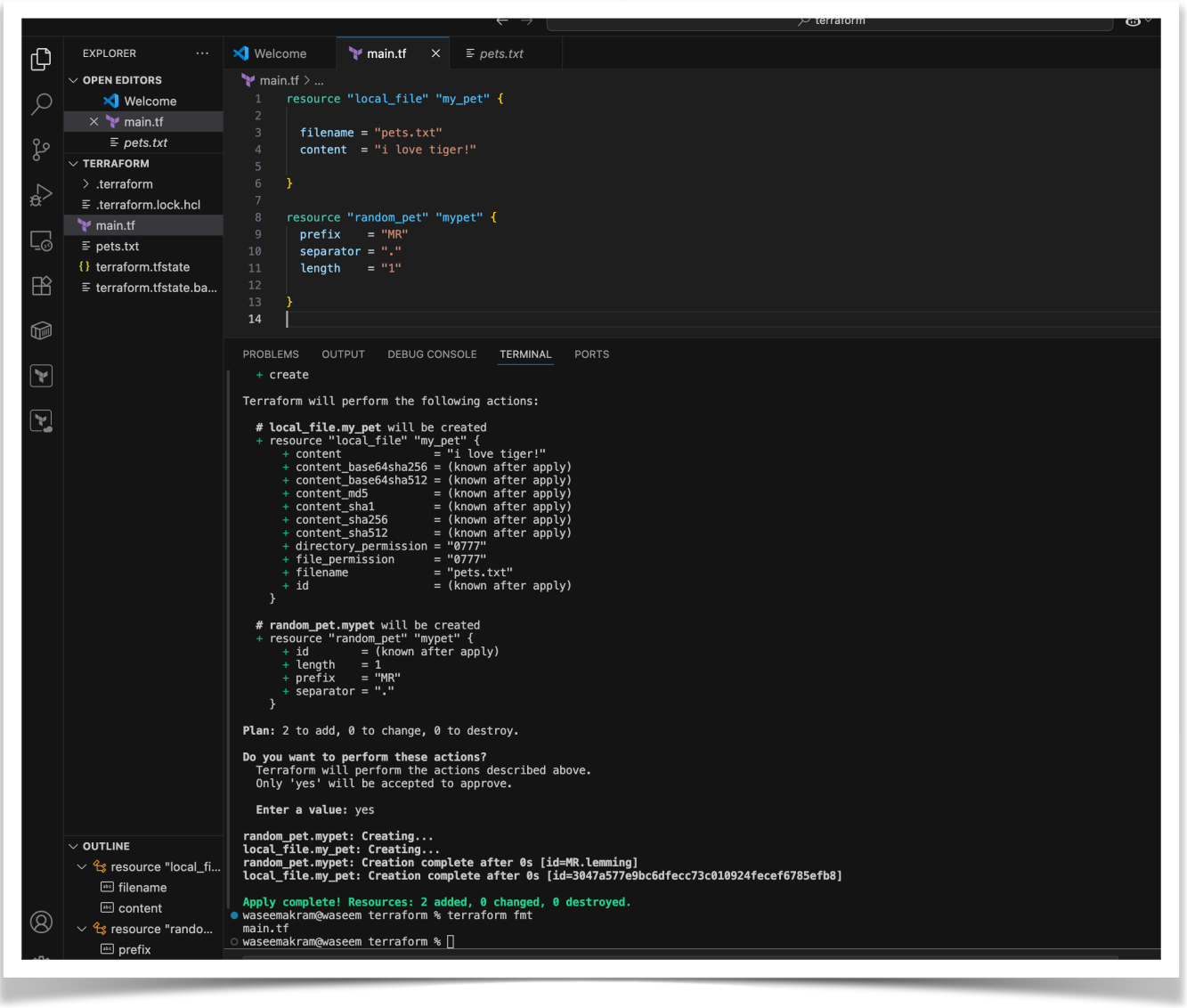
apply



```
resource "local_file" "my_pet" {

  filename = "pets.txt"
  content  = "i love tiger!"

}

resource "random_pet" "mypet" {
  prefix    = "MR"
  separator = "."
  length    = "1"

}
```






EXPLORER
OPEN EDITORS
Welcome
main.tf
pets.txt
TERRAFORM
.terraform
.terraform.lock.hcl
main.tf
pets.txt
terraform.tfstate
terraform.tfstate.ba...
OUTLINE
resource "local_fi...
filename
content
resource "rando...
prefix
Welcome
main.tf
pets.txt
main.tf > ...
1 resource "local_file" "my_pet" {
2
3 filename = "pets.txt"
4 content = "i love tiger!"
5
6 }
7
8 resource "random_pet" "mypet" {
9 prefix = "MR"
10 separator = "."
11 length = "1"
12
13 }
14
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
+ create
Terraform will perform the following actions:
# local_file.my_pet will be created
+ resource "local_file" "my_pet" {
+ content = "i love tiger!"
+ content_base64sha256 = (known after apply)
+ content_base64sha512 = (known after apply)
+ content_md5 = (known after apply)
+ content_sha1 = (known after apply)
+ content_sha256 = (known after apply)
+ content_sha512 = (known after apply)
+ directory_permission = "0777"
+ file_permission = "0777"
+ filename = "pets.txt"
+ id = (known after apply)
}
# random_pet.mypet will be created
+ resource "random_pet" "mypet" {
+ id = (known after apply)
+ length = 1
+ prefix = "MR"
+ separator = "."
}
Plan: 2 to add, 0 to change, 0 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
random_pet.mypet: Creating...
local_file.my_pet: Creating...
random_pet.mypet: Creation complete after 0s [id=MR.lemming]
local_file.my_pet: Creation complete after 0s [id=3047a577e9bc6dfecc73c010924fecef6785efb8]
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
waseemakram@waseem terraform % terraform fmt
main.tf
waseemakram@waseem terraform %

Task : 3

3) Note down below points,

Terraform Core Concepts

3.1 Terraform Init

Terraform Init is the first command that must be executed in any Terraform project.

Purpose:

- Initializes a Terraform working directory
- Downloads required provider plugins
- Creates the `.terraform` directory

Command:

terraform init

3.2 Terraform Plan

Terraform Plan is used to preview the changes Terraform will make.

Purpose:

- Shows execution plan
- Does not create or modify resources
- Helps in reviewing changes before applying

Command:

terraform plan

3.3 Terraform Apply

Terraform Apply is used to create or modify infrastructure.

Purpose:

- Executes the actions proposed in the plan

- Creates cloud resources
- Requires confirmation before execution

Command:

terraform apply

3.4 Terraform Provider

A Terraform Provider is responsible for interacting with cloud platforms.

Purpose:

- Defines the cloud service (AWS, Azure, GCP)
- Enables Terraform to communicate with cloud APIs

Example (AWS Provider):

provider "aws" {

region = “us-east-1"

}

Task :4

4).Integrate a sample Terraform template in Jenkins.

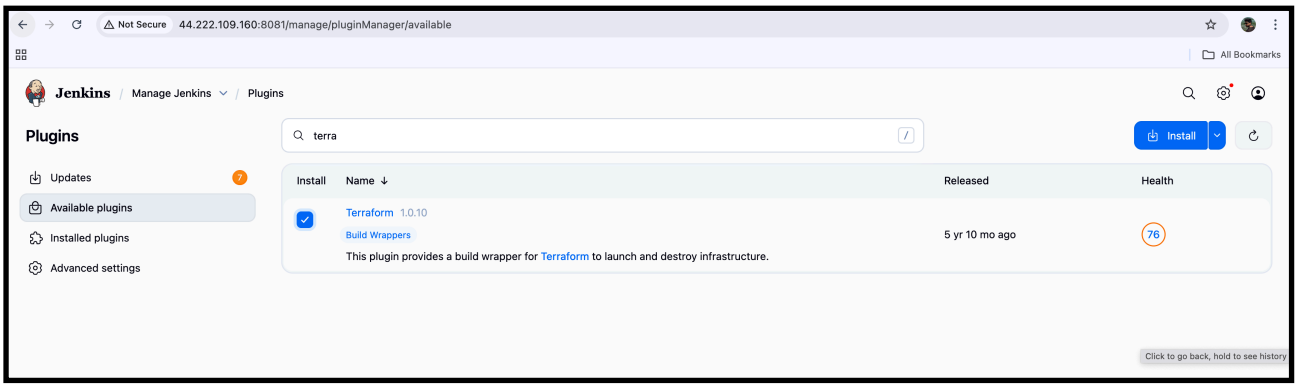
Objective :

To integrate a sample Terraform template with Jenkins so that Terraform commands can be executed automatically using a Jenkins pipeline.

Prerequisites :

1. **Jenkins server** up and running.
2. **Terraform installed** on Jenkins server (terraform -v should work).
3. Git is installed on Jenkins server
   -
4. A **Git repo** containing Terraform code (for example: main.tf). And jenkisfile

Install terraform plugin



Step-1 : In Jenkins server install terraform

wget https://releases.hashicorp.com/terraform/1.14.3/terraform_1.14.3_linux_amd64.zip

```
[root@ip-172-31-65-24 ~]# wget https://releases.hashicorp.com/terraform/1.14.3/terraform_1.14.3_linux_amd64.zip
--2026-01-08 11:40:05--  https://releases.hashicorp.com/terraform/1.14.3/terraform_1.14.3_linux_amd64.zip
Resolving releases.hashicorp.com (releases.hashicorp.com)... 3.171.85.65, 3.171.85.107, 3.171.85.80, ...
Connecting to releases.hashicorp.com (releases.hashicorp.com)|3.171.85.65|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 30895477 (29M) [application/zip]
Saving to: 'terraform_1.14.3_linux_amd64.zip'

terraform_1.14.3_linux_amd64.zip              100%[=========================================================================================>]

2026-01-08 11:40:06 (538 MB/s) - 'terraform_1.14.3_linux_amd64.zip' saved [30895477/30895477]

[root@ip-172-31-65-24 ~]#
```

Unzip terraform_1.14.3_linux_amd64.zip

```
[root@ip-172-31-65-24 ~]# unzip terraform_1.14.3_linux_amd64.zip
Archive:  terraform_1.14.3_linux_amd64.zip
  inflating: LICENSE.txt
  inflating: terraform
[root@ip-172-31-65-24 ~]#
```

sudo mv terraform /usr/local/bin/

Terraform -version

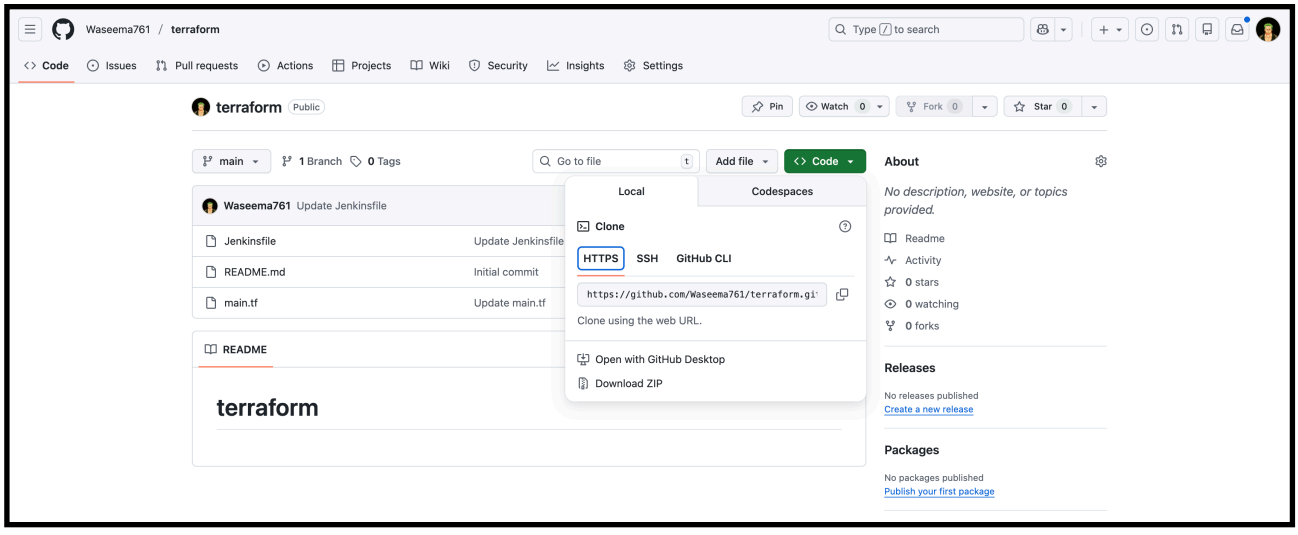
```
Archive:  terraform_1.14.3_linux_amd64.zip
  inflating: LICENSE.txt
  inflating: terraform
[root@ip-172-31-65-24 ~]# sudo mv terraform /usr/local/bin/
[root@ip-172-31-65-24 ~]# terraform -version
Terraform v1.14.3
on linux_amd64
[root@ip-172-31-65-24 ~]#
```

step 2:

In Jenkins tools added terrform

Step 3:

Create a New repo in git hub and and 2 new files main.tf and Jenkinfile in the git hub repo



github.com/Waseema761/terraform/blob/main/Jenkinsfile

Waseema761 / terraform

Code Issues Pull requests Actions Projects Wiki Security Ins

main terraform / Jenkinsfile

Waseema761 Update Jenkinsfile

Code Blame 30 lines (25 loc) · 514 Bytes

```
pipeline {
    agent any

    stages {
        stage('Checkout SCM') {
            steps {
                checkout scm
            }
        }

        stage('Terraform Init') {
            steps {
                sh 'terraform init'
            }
        }

        stage('Terraform Plan') {
            steps {
                sh 'terraform plan'
            }
        }

        stage('Terraform Apply') {
            steps {
                sh 'terraform apply -auto-approve'
            }
        }
    }
}
```

github.com/Waseema761/terraform/blob/main/main.tf

Code Issues Pull requests Actions Projects Wiki Security Insights

main terraform / main.tf

Waseema761 Update main.tf

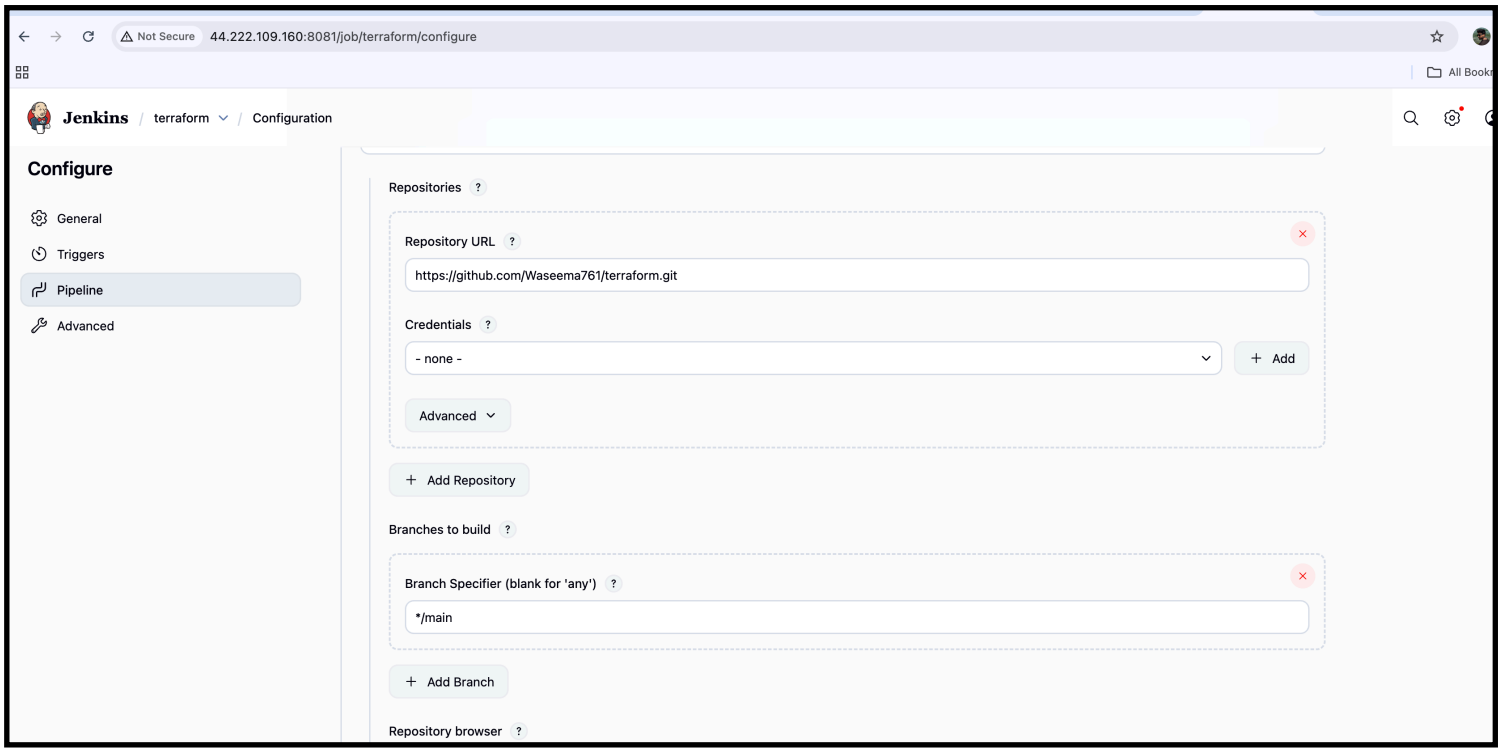
Code Blame 5 lines (4 loc) · 89 Bytes

```
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "i love tiger!"
}
```

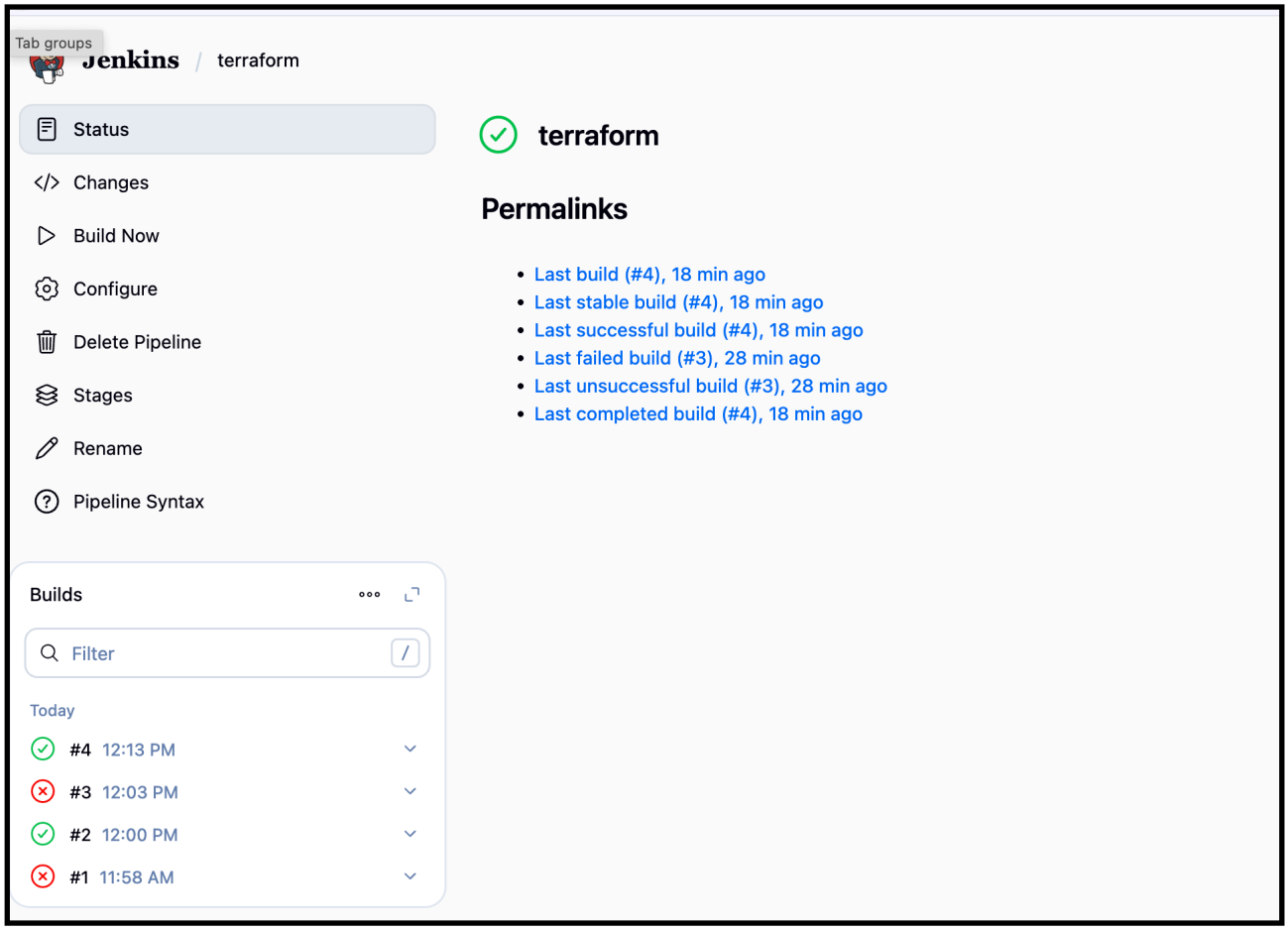
Step 4:

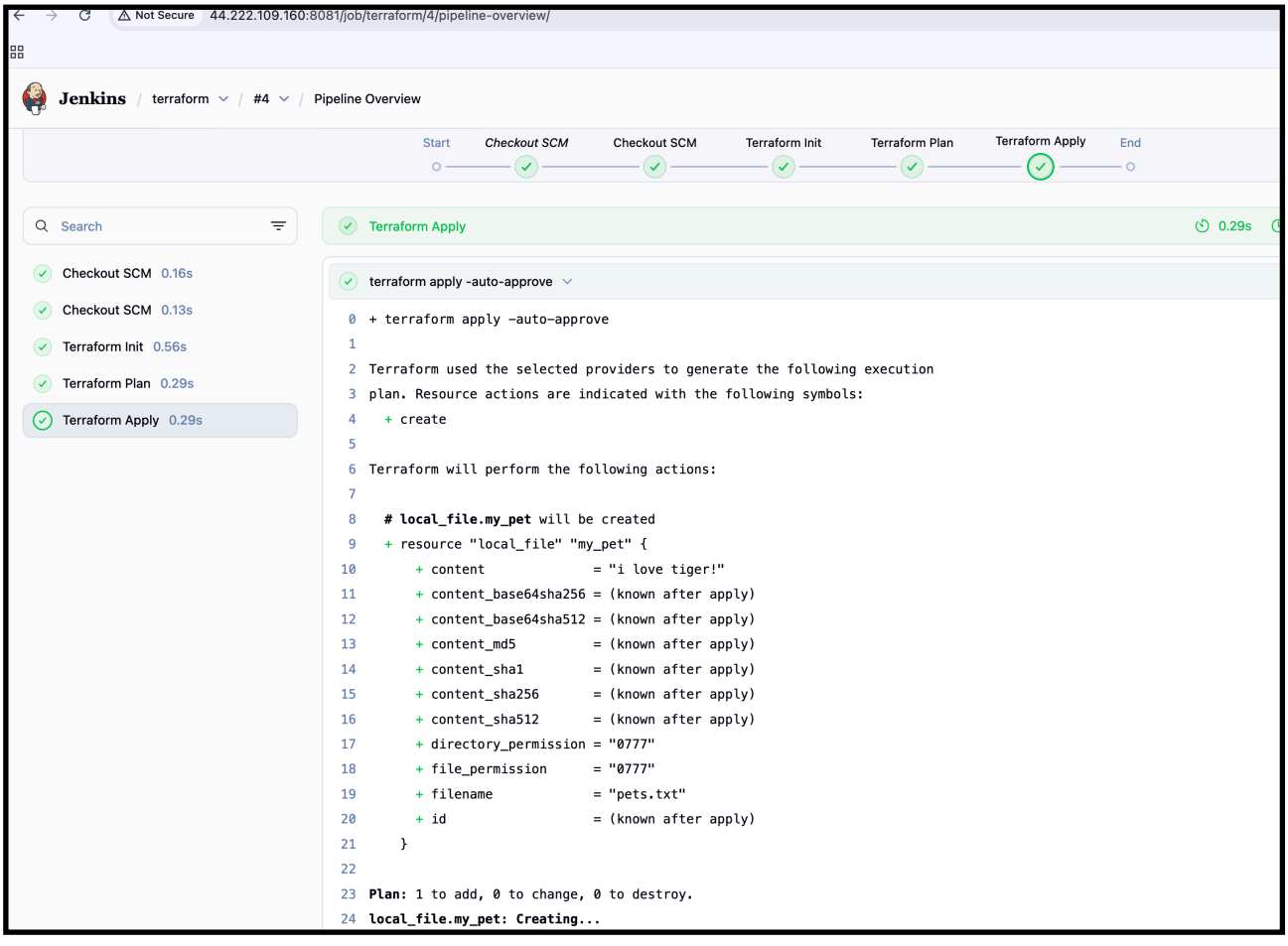
Create a new item terraform job.in Jenkins

And select the pipeline and the git hub url and the branch in the configurations



—-> save the and build


Tab groups
Jenkins / terraform
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
terraform
Permalinks
Last build (#4), 18 min ago
Last stable build (#4), 18 min ago
Last successful build (#4), 18 min ago
Last failed build (#3), 28 min ago
Last unsuccessful build (#3), 28 min ago
Last completed build (#4), 18 min ago
Builds
Filter
Today
#4 12:13 PM
#3 12:03 PM
#2 12:00 PM
#1 11:58 AM


Not Secure 44.222.109.160:8081/job/terraform/4/pipeline-overview/
Jenkins / terraform / #4 / Pipeline Overview
Start
Checkout SCM
Checkout SCM
Terraform Init
Terraform Plan
Terraform Apply
End
Search
Checkout SCM 0.16s
Checkout SCM 0.13s
Terraform Init 0.56s
Terraform Plan 0.29s
Terraform Apply 0.29s
Terraform Apply
0.29s
terraform apply -auto-approve
+ terraform apply -auto-approve
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
# local_file.my_pet will be created
+ resource "local_file" "my_pet" {
+ content = "i love tiger!"
+ content_base64sha256 = (known after apply)
+ content_base64sha512 = (known after apply)
+ content_md5 = (known after apply)
+ content_sha1 = (known after apply)
+ content_sha256 = (known after apply)
+ content_sha512 = (known after apply)
+ directory_permission = "0777"
+ file_permission = "0777"
+ filename = "pets.txt"
+ id = (known after apply)
}
Plan: 1 to add, 0 to change, 0 to destroy.
local_file.my_pet: Creating...

Verification :

Now lets check in Jenkins workspace

```
[root@ip-172-31-65-24 jenkins]# cd workspace/
[root@ip-172-31-65-24 workspace]# ls
'MULTI-STAGE PIPELINE'       hiring-app-declarative       job2                    pipeline      'scripted pipeline'       simplecustomerapp-pipeline       vprofile
'MULTI-STAGE PIPELINE@tmp'   hiring-app-declarative@tmp   job3                    pipeline@tmp  'scripted pipeline@tmp'   simplecustomerapp-pipeline@tmp
 first-job                   hiring-app-parallel          parametirized-job       private       scripted-pipeline1        terraform
 hiring-app                  hiring-app-parallel@tmp      parametirized-job@tmp   private@tmp   scripted-pipeline1@tmp    terraform@tmp
[root@ip-172-31-65-24 workspace]# cd terraform
[root@ip-172-31-65-24 terraform]# ls
Jenkinsfile  README.md  main.tf  pets.txt  terraform.tfstate
[root@ip-172-31-65-24 terraform]#
```

We are able to see the pets.txt file

—> now to confirm lets cat the file

```
Jenkinsfile  README.md  main.tf  pets.txt  ter
[root@ip-172-31-65-24 terraform]# cat pets.txt
i love tiger![root@ip-172-31-65-24 terraform]#
```

```
[root@ip-172-31-65-24 terraform]# cat pets.txt
i love tiger!
```

—> same as given below code from GitHub